

JAMES RUSE AGRICULTURAL HIGH SCHOOL
1998 4 UNIT MATHEMATICS
TERM 4 ASSESSMENT

Time: 85 minutes.

No equipment may be borrowed during the examination.

All necessary working must be shown.

Question 1

- (a) If $z = 2 - 3i$ and $\omega = -1 - i$, find
- (i) $|z|$
 - (ii) $\arg \omega$
 - (iii) $\omega \bar{z}$
- (b) Find $\sqrt{-5 - 12i}$ in the form $a + ib$.
- (c) If $z = 1$ is one of the roots of the equation $iz^2 + (ia - 1)z + (i - b) = 0$, where a and b are real, find:
- (i) the values of a and b .
 - (ii) the other root.
- (d) If $z = 4\lambda + 3i(1 - \lambda)$ and λ is real,
- (i) prove that the point P which represents z on the Argand plane lies on a straight line.
 - (ii) find the least value of $|z|$.

Question 2 (Start a new page)

- (a) Express in mod-arg form:
- (i) $\frac{5 + i}{2 + 3i}$
 - (ii) $i\bar{z}$ if $z = 4(\cos \theta + i \sin \theta)$.
- (b) Find all n for which $(2 - 2\sqrt{3}i)^n$ is real.
- (c) What single transformation maps z to:
- (i) $\bar{z}$
 - (ii) iz
- (d) Show that there is only one value of z for which both $\arg(z + i) = \frac{\pi}{4}$ and $|z + 1| = 2$, and that this value is real.

Question 3 (Start a new page)

- (a) Given $z = x + iy$ and $P(x,y)$ moves in the complex plane, graph the locus of P and state its Cartesian equation if:
- $|z - i| = |z + 1|$
 - $\arg\left(\frac{z+2}{z-2}\right) = \frac{\pi}{2}$
 - $\operatorname{Im}\left(z + \frac{1}{z}\right) = 0$
- (b) Using suitable diagram(s) on the Argand plane, prove the following inequalities from **geometric considerations** alone. Give clear reasons.
- $|z_1 + z_2| \leq |z_1| + |z_2|$.
 - $|z - 1| \leq ||z| - 1| + |z| |\arg z|$, where $-\pi < \arg z \leq \pi$.

Question 4 (Start a new page)

- (a) If $\alpha = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$, $\beta = \alpha + \alpha^2 + \alpha^4$ and $\gamma = \alpha^3 + \alpha^5 + \alpha^6$,
- prove that $\beta + \gamma = -1$ and $\beta\gamma = 2$.
 - Hence show that: $\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} = \frac{\sqrt{7}}{2}$.
- (b) In the Argand plane, consider the triangle ABC with vertices at $A(-1,0)$, $B(1,0)$ and C represents the complex number z , for $-\pi < \arg z \leq \pi$.
Let L be a straight line through the origin and parallel to the bisector of $\angle ACB$.
Prove that both values of $\sqrt{z^2 - 1}$ lie on the line L , clearly labelling all important features in your diagram.
- (c) The points $P_1, P_2, P_3, \dots, P_n$ representing the complex numbers $z_1, z_2, z_3, \dots, z_n$ lie below the straight line $y = \sqrt{3}x$.
- The points $Q_1, Q_2, Q_3, \dots, Q_n$ representing $\frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3}, \dots, \frac{1}{z_n}$ possess a similar property. State this property and the equation of the corresponding straight line.
 - Show that $z_1 + z_2 + \dots + z_n \neq 0$, for **any** set of n points, and similarly $\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} + \dots + \frac{1}{z_n} \neq 0$.

END of PAPER

QUESTION 1.

[4]

a) (i) $|z| = \sqrt{2^2 + 3^2} = \sqrt{13}$ (1)

(ii) $\arg w = -\frac{3\pi}{4}$ (1)

(iii) $w\bar{z} = (-1-i)(2+3i)$
 $= -2-3i-2i+3$
 $= 1-5i$ (2)

[3]

b) $\sqrt{-5-12i} = x+iy$
 $-5-12i = (x+iy)^2$
 $\begin{cases} x^2 - y^2 = -5 \\ 2xy = -12 \end{cases} \Rightarrow y = \frac{-6}{x}$ (3)

$\therefore x^2 - \frac{36}{x^2} = -5$
 $x^4 + 5x^2 - 36 = 0$
 $\therefore x = 2 \text{ or } x = -2$
 $y = -3 \text{ or } y = 3$
 $\therefore \text{roots } 2-3i, -2+3i$ (1)

[5]

c) (i) $i + ia - 1 + i - b = 0$ (1)
 $\begin{cases} -1-b = 0 \\ 2+a = 0 \end{cases} \therefore b = -1$ (3)
 $\underline{a = -2}$ (1)

(ii) $\alpha + \beta = -a + \frac{1}{i} = 2 - i$ (2)
 $\therefore \alpha = 1$ $\therefore$ other root $\underline{1-i}$ (1)

[3]

d) (i) $z = 4\lambda + 3i(1-\lambda)$ where $z = x+iy$
 $\begin{cases} x = 4\lambda \\ y = 3(1-\lambda) \end{cases} \lambda = \frac{x}{4}$ (2)
 $y = 3 - \frac{3x}{4}$ (1)

$\therefore P$ lies on the straight line $\underline{3x+4y=12}$

(ii) $d = \frac{|10+0-12|}{\sqrt{3^2+4^2}} = \underline{\underline{2\frac{2}{5}}}$ (1)

QUESTION 2.

[6]

a) (i) $\frac{5+i}{2+3i} = \frac{(5+i)(2-3i)}{(2+3i)(2-3i)} = \frac{13-13i}{13} = 1-i$ (1)

(ii) $i\bar{z} = i(4\cos\theta - 4i\sin\theta) = 4(\sin\theta + i\cos\theta)$ (3)
 $= 4\left(\cos\left(\frac{\pi}{2} + \theta\right) + i\sin\left(\frac{\pi}{2} + \theta\right)\right)$ (1)

[5]

b) $(2-2\sqrt{3}i)^n = \left[4\left(\cos\frac{-\pi}{3} + i\sin\frac{-\pi}{3}\right)\right]^n$
 $= 4^n \left(\cos\frac{-n\pi}{3} + i\sin\frac{-n\pi}{3}\right)$ (3)

Real for $\sin\frac{-n\pi}{3} = 0$ (1)
 $\therefore \frac{n\pi}{3} = k\pi \therefore \underline{n = 3k, k \text{ integer}}$ (2)

[2]

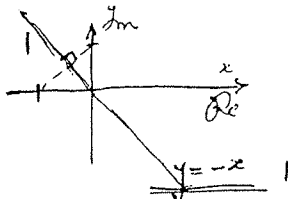
c) $z \rightarrow \bar{z}$ reflection on X axis (1)
 $iz \rightarrow i\bar{z}$ rotation of $\frac{\pi}{2}$, anticlockwise (2)

d) $\begin{cases} \arg(z+i) = \frac{\pi}{4} \\ |z+i| = 2 \end{cases} \Rightarrow y = x-1, x > 0$ (1)
 $\Rightarrow (x+1)^2 + y^2 = 4$ (1)
 $\therefore 2x^2 + 2 = 4 \therefore x = \pm 1$ (4)
 but $x > 0 \therefore \underline{x = 1}$ is the only solution
 $\therefore \underline{z = 1}$ (1)

[4] if from
 diagram
 give 4 marks

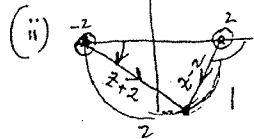
QUESTION 3.

a) (i)



$$|z-i| = |z+i|$$

(2)



$$\arg\left(\frac{z+2}{z-2}\right) = \frac{\pi}{2}$$

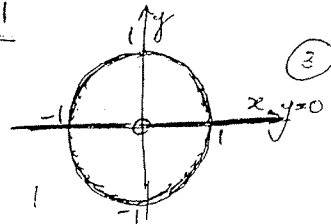
$$x^2 + y^2 = 4, y < 0$$

(3)

$$\text{Im}\left(z + \frac{1}{z}\right) = 0$$

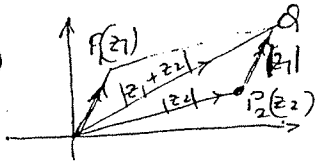
Either z real $\neq 0$ or $\frac{1}{z} = -\bar{z}$

$$\therefore y = 0, x \neq 0, \quad x^2 + y^2 = 1$$



(3)

b) (i)



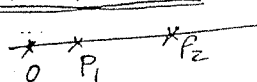
OP_1, OP_2 from by construction

In $\triangle OP_1P_2$ $OP_1 < OP_2 + P_1P_2$

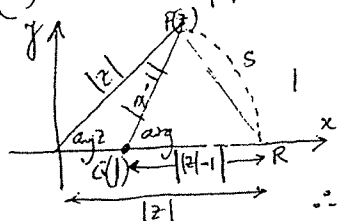
"each side of a $\triangle$ is less than the sum of the other two sides"

$$\therefore |z_1 + z_2| < |z_1| + |z_2|$$

But if OP_1, P_2 are in a straight line then the equality stands $|z_1 + z_2| = |z_1| + |z_2|$



(ii) Case 1. $|z| > 1$



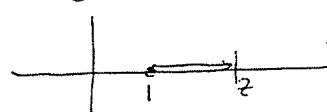
$$|z-1| < |z|-1 + |PR|$$

but arc $BR = |z| \arg z$
 $l = r\theta, |PR| < \text{arc}$

$$\therefore |z-1| < |z|-1 + |z| \arg z$$

(4)

Case 2. z real with $\theta = 0$



$$\therefore |z-1| = ||z|-1|$$

$\therefore$ equality applies since $|\arg z| = 0$

$$|z-1| = ||z|-1| + |z| |\arg z|$$

Bonus mark if student considers $|z| < 1$ and $|z| = 1$ chord less

QUESTION 4.

a)

$$z^7 = 1 = \cos 2n\pi + i \sin 2n\pi \Rightarrow x = \cos \frac{2n\pi}{7} + i \sin \frac{2n\pi}{7}$$

$$\rho = \begin{pmatrix} \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \\ \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} \\ \cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7} \end{pmatrix} = \begin{pmatrix} \cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7} \\ \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} \\ \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \end{pmatrix} \quad x^7 = 1$$

$$\beta + \gamma = x + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 = (x^7 - 1) = 0$$

(2)

[7]

$$\beta \gamma = (x + x^2 + x^3)(x^3 + x^5 + x^6) = x^4 + x^6 + 1 + x^5 + 1 + x + 1 + x^2 + x^3 = x + x^2 + x^3 + x^4 + x^5 + x^6 + 3 = -1 + 3 = 2$$

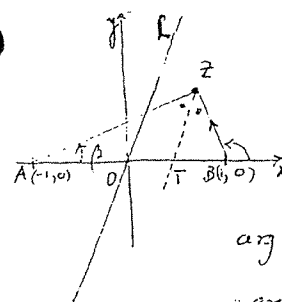
$$\text{(ii)} \quad \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{6\pi}{7} = \text{Im}(\beta)$$

$$\text{To find } \beta \text{ of } x^2 + x + 2 = 0 \therefore x = \frac{-1 \pm \sqrt{7}i}{2}$$

$$\therefore \beta = \frac{-1 + \sqrt{7}i}{2}, \quad +\sqrt{7} \text{ since } \sin \frac{4\pi}{7} > \sin \frac{6\pi}{7}$$

$$\therefore \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{6\pi}{7} = \frac{\sqrt{7}}{2}$$

b)



$$\arg(z-1) = \alpha, \quad \arg(z+1) = \beta$$

$$\angle TCB = \frac{\alpha - \beta}{2} \therefore \angle TCB = \frac{\alpha + \beta}{2}$$

$$\therefore \angle LOX = \frac{\alpha + \beta}{2}$$

$$\arg \sqrt{z^2 - 1} = \frac{1}{2} [\arg(z-1) + \arg(z+1)]$$

$$\therefore \arg(\sqrt{z^2 - 1}) = \frac{1}{2} (\alpha + \beta) = \angle LOX$$

$\therefore$ both values of $\sqrt{z^2 - 1}$ lie on line L.

c) (i) Since $\arg \frac{1}{z} = -\arg z$ the points representing $\frac{1}{z_1}, \frac{1}{z_2}, \dots, \frac{1}{z_n}$ lie above the line $y = -\sqrt{3}x$

(ii) By rotating the full plane $\frac{\pi}{6}$ anticlockwise, all points will have $x > 0$ $\therefore$ let $w = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$

$$\therefore w z_1 + w z_2 + \dots + w z_n \neq 0 \quad \text{since } \text{Re}(w z_i) > 0$$

$$\therefore \text{Re}(w z_1 + w z_2 + \dots + w z_n) > 0$$

$$\therefore w(z_1 + z_2 + \dots + z_n) \neq 0 \quad \text{since } z = 0 \text{ iff } x = 0$$

$$\therefore z_1 + z_2 + \dots + z_n \neq 0$$

Similarly for $L \perp \dots \perp$ with $w = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$